

Caffevolve — Phone-Primary Personalization

Addendum to “The Intelligence Center as a Distributed Process”: how resident profiles are reserved, finished, paired and kept in sync

Prepared for: Brent Langevin **Purpose:** Define which personalization tasks happen on the phone and which at the device, how a resident profile is reserved and paired, how edits stay in sync, and the commitments that keep the device useful on its own. **Status:** working draft, revised after review. The rules below were agreed in review; section 14 lists what is still open.

1. The core rule, in one line

The phone is the authoring surface; the device is the execution surface. Heavy personalization is done where the user is comfortable, and both surfaces may adjust what exists. All brew, grind and drink-path decisions, and every brew initiation, remain at the device.

2. Why

Personalization has grown: the initial profile, the Caffeine Behavior Instrument (CBI), subprofiles, BrewNow profiles, additive and grind preferences, and now daily plans and routines. Doing all of it at the device means long sessions standing at a countertop. Moving the heavy, optional authoring to the phone removes that friction and should reduce reluctance to use the personalization tools. The architecture already supports it: the engine logic is compiled into the app, and the per-user cloud store already carries the event log and the user’s parameters between the two copies of the engine.

3. Two stages of a resident profile

A Resident User Profile is built in two stages, by two different people.

Stage	Who	Where	What it holds
Basic Resident Profile	The administrator, behind the Admin PIN	At the device (Administrator Functions)	First name, initials, profile color and the user’s phone number . It reserves one of the device’s profile slots.
Profile setup, producing the Complete Resident Profile	The user, with no administrator PIN	Device or phone	The CBI and the ceilings it derives, then the optional layer described in section 5.

- **Every resident profile starts this way, including the administrator’s own.** The administrator first sets up the device (connection, beans, additives and the Admin PIN). No

profile exists yet. The administrator then creates their own Basic Resident Profile, at that time or later, by the same route as everyone else.

- **The administrator never answers the CBI for anyone.** The slot is reserved by the administrator; everything personal is completed by the user.
- **Until the user finishes,** the profile appears as “Ready for you” with a “Finish setup” action, and the resident brewing paths (Home, Away, Plan my Day) stay locked. The user can brew through any guest channel in the meantime. The user may pause and resume setup, but cannot discard the reserved slot. Only the administrator removes a resident profile.
- **Resumable.** Setup already offers “Exit setup & complete later.” A user may answer part of the CBI on one surface and finish on either. The profile is established only when the CBI completes.
- **Bedtime is not part of the profile.** It is a session value, set and revisited at the device with each brew. It is never synced and cannot conflict.
- **Slots** are limited by tier (Originara 2, Advancara 5, Culminara Office 20), so the administrator decides who holds one.

4. Reserving, inviting and pairing

- **Phone number is part of the administrator’s step.** It is entered with the name, initials and color. It is a delivery address for the invite code; it is not a credential.
- **The code.** On completing the Basic Resident Profile, the device registers the shell (name and phone number) with the cloud and requests a one-time code, which is texted to that number. The device must be online at that moment. The user’s personal data (CBI answers, ceilings, preferences) stays on the device until they pair.
- **The code is one-time, expires after a short window, locks after a few wrong attempts, and binds to the first account that redeems it.** The administrator can issue a fresh code if one expires or is lost.
- **Cleanup.** If a code expires unused, the phone number is deleted from the cloud. After the code is redeemed, the number is deleted from the cloud as well, unless it proves to be needed for account sign-in (section 14). The device keeps its own copy in the profile.
- **A profile is bound to one account permanently.** The claim cannot be undone or transferred, and there is no reassign function. If the wrong person claims a profile, for example because the number was mistyped, the administrator removes the profile and creates a new Basic Resident Profile. Removal deletes the cloud copy and unpairs the account.
- **Changing the number.** The user can edit their phone number on the profile review screen, on either surface. The change is confirmed with a code sent to the new number, and the old number stays until it is entered.
- **A new phone** is handled by signing into the account on it, not by the device or the administrator (the sign-in method is open, section 14).

- **An unpaired profile works fully at the device.** A local-only profile has no cloud backup, and a device reset or failure loses it. The app and the device say so when offering the invite.
- **Phone-first.** A person cannot self-create a resident profile on a household device. The administrator-issued invite is the only route to a slot, so one mechanism covers both directions.

5. What is created where

Item	Create	Edit
Basic Resident Profile: name, initials, color, phone number	Administrator, at the device (Admin PIN)	Phone number: the user, on either surface. Other fields as today.
Complete Resident Profile: CBI and the ceilings it derives	User, device or phone. Phone preferred. Resumable.	Either, through the profile review screen
Optional layer: additive preferences, grind setting, subprofiles, BrewNow profile, workplace settings	Phone only	Either
Plans and routines (Plan my Day)	Phone only	Phone (full). Device: skip today's plan or make light edits; Plan Check runs at the device.
Guest profile (Quick Profile Path)	Device only	Device only

- **Complete on the device.** A resident can finish the CBI and ceilings at the device with no phone and no account. The device is never unusable for want of a phone. The device first-run screen says the longer setup is faster on the phone and offers to continue there.
- **Phone-only layer.** On the device, screens for the optional layer say “available in the CaffeolveGo app” and do not appear broken. Until the user sets them, defaults apply (open item, section 14).
- Each item is created in one place at a time, which removes most conflict risk. Remaining conflicts occur only on edits, covered in sections 7 and 8.

6. The boundary model

The Caffeine Comfort Zone has four parameters. Users can adjust only the two ceilings.

Parameter	Source	User-adjustable
Wake-Hours Ceiling	CBI, then refined by the AFS	Yes, any time, from the profile review screen (“Fine-tune ceilings”)
Bedtime Ceiling	CBI, then refined by the AFS	Yes, same as above
Peak-Zone Floor	CBI, then refined by the AFS	No

Parameter	Source	User-adjustable
Caffeine Decay Rate	CBI, then refined by the AFS	No

- **Each ceiling has three parts.** The *computed value* (from the CBI, moved by the AFS); an optional *user override* (set with the slider); and the *effective value*, which is the override if one exists, otherwise the computed value. The device brews against the effective value. “Use computed” removes the override.
- **The platform only advises.** Setting a ceiling higher than most people are comfortable with shows an advisory message, not a block. The user decides.
- **The AFS never overwrites an override.** It continues to refine the computed value in the background, so “Use computed” returns the learned value. Its boundary-triggered prompts fire around the effective value.
- **Refinement is stored as feedback events.** Each feedback response is an immutable event with a source and ID, and the computed value is derived from the baseline plus those events. Events arrive from either surface and merge as a simple union, in any order, so the two surfaces cannot disagree about the computed value once they hold the same events and the same engine version.

7. Editing and conflicts

The existing principle carries over: state changes are appended, not edited in place. Applied to profiles, subprofiles, preferences, the phone number and plans:

- Every edit produces a **new version of the record**, stamped with its source (phone or device), time and the version it was based on.
- If both surfaces edited from the same base version, the **later edit wins per field**. A ceiling change on one side and a rename on the other both survive.
- If both changed the **same field**, the later edit wins and the other surface shows a one-line notice that its earlier change was replaced.
- The **two ceilings are treated as a group**, because they are adjusted together. If both surfaces edited them, the later edit wins for the pair, so the result is never a mix the user did not see.
- Each surface shows **sync state** (for example, “Synced to your Caffevolve 2 min ago”), especially after a ceiling change, because the device brews against whatever it last received.

For brew events the device remains the source of truth. For personalization records, the most recent valid version is authoritative, whichever surface wrote it.

8. Edit Profile: re-running setup

From the profile review screen, **Edit Profile** re-runs the whole setup flow, and everything in it can be changed: the CBI answers, subprofiles, grind setting and additive preferences. It is

available on either surface. A re-run has two kinds of outcome:

- **CBI answers unchanged.** Only the other fields change, and they merge as ordinary field edits. The user's ceiling overrides and the AFS learning are kept. Changing an additive must not wipe tuned ceilings.
- **CBI answers changed.** This is a new baseline. The computed boundaries and decay rate are recomputed, the user's **ceiling overrides are cleared**, and the **AFS learning starts over**. The user passes through the fine-tune screen again and may set new overrides.
- Any edit made on the other surface against the old baseline is superseded by the new baseline, and that surface shows a notice.

9. Guests

The phone-first rules apply to Resident Users only. The Guest path stays entirely on the device: name, initials, self-selected sensitivity level and bedtime, with no account, no invite and no CBI. A Returning Guest Profile continues to be kept for 72 hours.

10. Safety and access

- **The Admin PIN stays, and is unchanged in purpose.** It secures the device and decides who holds a profile on it. It covers device configuration (network, beans, canisters, hardware), reserving a slot, issuing an invite code, removing a resident profile, and resetting or transferring the device (which wipes local profiles and unpairs the device). One entry per visit to Administrator Functions.
- **The profile PIN** exists only on the shared Culminara Office device: a user-chosen four-digit PIN required when a profile is selected. The home device carries none. The two PINs protect different things and never overlap.
- **Profile edits need no extra PIN.** Adjusting a ceiling, changing additives or re-running setup follows the profile's own rule: nothing at home, the profile PIN on the Office device.
- The platform continues only to advise. The user decides what to brew and what their ceilings are.

11. Design commitments

- **Minimal on-device path.** The device always supports the Complete Resident Profile (CBI and ceilings) and the Guest path, so it works without a phone.
- **Full local copy.** The device holds a complete copy of the active profiles and plans, so a cloud or app outage never strands a user's setup. Brewing continues from the last synced state.
- **The cloud remains a store, not a brain.** It holds the record and the current versions. It does not compute residual or decide what to brew.

- **Data governance defined before launch.** CBI answers describe sleep and caffeine habits and are health-adjacent, and the phone number is personal data entered by one person about another. Consent, retention, export and account deletion rules need to be set, in addition to per-user isolation and encryption.
- **Version matching extended.** The requirement that firmware and app carry the same engine version also covers the record schema and the shared personalization flows, so older firmware does not meet newer app flows unprepared.
- **Validate the assumption.** Test completion rates for the CBI and planning on a phone prototype against the device before committing the full build.

12. Prototype and reference changes this requires

- Add a **phone number field** to the administrator’s “Set Up a New User” step and to the first-profile step.
- Add an **invite-code step** after the Basic Resident Profile is created, with an on-screen confirmation that a code was sent.
- Add an editable **phone number** to the profile review screen, with confirmation by code.
- Change the device setup flow so the optional layer (additive, grind, subprofiles, BrewNow) is created in the app, with an “available in the CaffevolveGo app” message at the device.
- Reconcile the reference text: the first profile is created by the administrator like any other, and the profile PIN applies to the Office tier only.
- Add the app-side screens for pairing, profile setup and the review screen.

13. Decisions recorded in review

- A profile is never moved to a different account.
- No profile PIN on home devices; the Admin PIN stays.
- The ceiling overrides and the AFS reset only when the CBI answers change.
- The AFS never overwrites a user’s override.

14. Open items

- **Account sign-in and recovery.** How a person signs into their own account on a new or second phone, and what happens when access is lost. This also decides whether the phone number is only a delivery address or also the sign-in identity, and so whether the cloud keeps it.
- Confirm that the Peak-Zone Floor is not user-adjustable (the prototype offers fine-tuning only for the two ceilings).

- The defaults that apply to additive and grind preferences until the user sets them on the phone.
- The exact set of light plan edits allowed at the device.
- Whether the Office-tier profile PIN also applies when a profile is edited from the phone.
- Plan my Day (FIG. 43A, step 3802) lists bedtime as loaded with the profile, but bedtime is a session value. Decide where planning gets its bedtime, particularly on the phone, and adjust the figure. Add a phone authoring entry point to FIG. 43A.
- Device resale and transfer: what happens to cloud-paired profiles.
- The prototype does not yet model AFS learning state, sync or pairing; these are specified here but not yet shown.

This addendum records the direction as agreed in review. Items in section 14 are not yet settled.